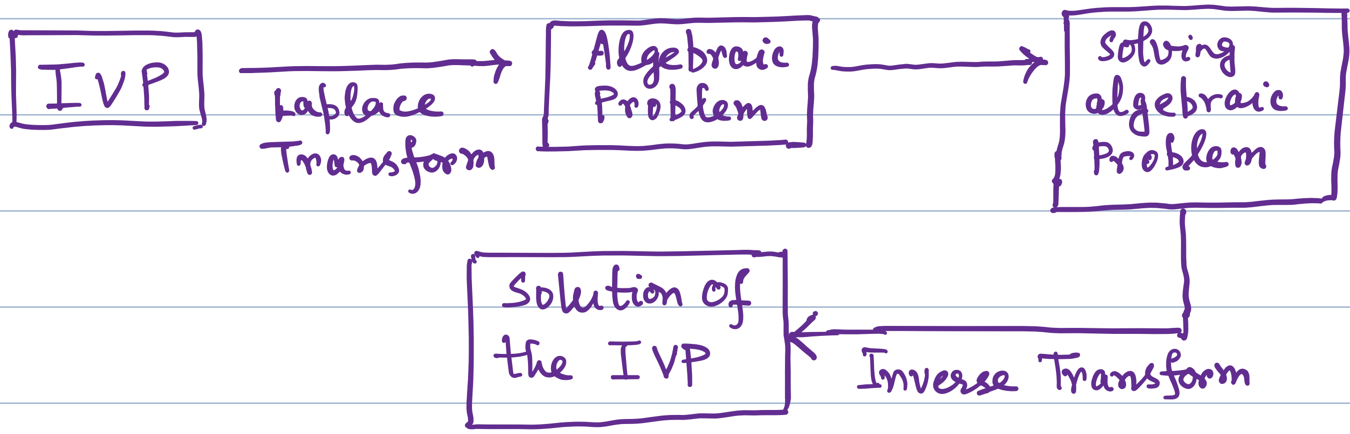
MTH 204 : Lecture 18

Laplace Transforms :

The process of solving an ODE using the Laplace transform method consists of three steps.



- The type of mathematics that converts problems of calculus to algebraic problems is known as Operational Calculus.

- <u>Advantages of the Method</u>:

(1) IVP can be solved without first determining a general solution. Nonhomogeneous ODEs are solved without first solving the corresponding homogeneous ODE.

(2) The use of unit step function (Heaviside

function and Dirac's Delta) make the method powerful for problems with inputs that have discontinuities or represent short impulses or complicated periodic functions.

Laplace Transform :

If $f(t)$ is a function defined for all $t \geq 0$, then its Laplace transform is defined by

$$F(s) = \mathcal{L}(f)(s) = \mathcal{L}(f) = \int_0^{\infty} e^{-st} f(t)\,dt \quad \text{------}(1)$$

We assume that $f(t)$ is such that the above integral exists (that is, has some finite value)

- The above is an improper integral and, by definition, is evaluated according to the rule

$$\int_0^{\infty} e^{-st} f(t)\,dt = \lim_{T \to \infty} \int_0^{T} e^{-st} f(t)\,dt$$

(provided the limit exists)

- The Laplace transform is an integral transform

$$F(s) = \int_0^{\infty} k(s,t) f(t)\,dt$$

with kernel $K(s,t) = e^{-st}$ and it transforms a function in one space to a function is another space.

In general the kernel is a function of the (two) variables in the two spaces.

- The given function $f(t)$ in (1) is called the inverse transform of $F(s)$ and is denoted by $\mathcal{L}^{-1}(F)$.

  We write $f(t) = \mathcal{L}^{-1}(F)$ ....(2)

From (1) and (2), we get $\mathcal{L}^{-1}(\mathcal{L}(f)) = f$ and
$\mathcal{L}(\mathcal{L}^{-1}(F)) = F$

<u>Notation:</u>

- original functions are denoted by lowercase letters and their transforms by the same letters in capital.
- Original functions depend on t and their transform on s

Ex: Calculate the Laplace transform of

(a) $f(t) = 1$ and (b) $f(t) = e^{at}$ when $t \geq 0$

(a) $F(s) = \int_0^\infty e^{-st} \, 1 \, dt = \left[\frac{e^{-st}}{-s} \right]_0^\infty = \boxed{\frac{1}{s}}$ $(s>0)$

(The above is really $\lim_{T \to \infty} \int_0^T e^{-st} \times 1 \, dt$

$= \lim_{T \to \infty} \left[\frac{e^{-st}}{-s} \right]_0^T$)

(b) $F(s) = \int_0^\infty e^{-st} e^{at} dt = \int_0^\infty e^{-(s-a)t} dt$

$= \left. \frac{e^{-(s-a)t}}{-(s-a)} \right|_0^\infty$

The above exists when $s-a>0$ i.e. $s>a$

Hence $F(s) = \mathcal{L}(e^{at})(s) = \frac{1}{s-a}$ for $s>a$

Linearity of Laplace Transform:

The Laplace transform is a linear operation: that is, for any functions $f(t)$ and $g(t)$

whose transforms exist and for any constants a and b, the transform of $af(t)+bg(t)$ exists and

$$\mathcal{L}\{af(t)+bg(t)\} = a\mathcal{L}\{f(t)\} + b\mathcal{L}\{g(t)\}$$

__Proof:__ $\mathcal{L}\{af(t)+bg(t)\}$

$$= \int_0^\infty e^{-st}[af(t)+bg(t)]\,dt$$

$$= a\int_0^\infty e^{-st} f(t)\,dt + b\int_0^\infty e^{-s\cdot t} g(t)\,dt$$

$$= a\mathcal{L}\{f(t)\} + b\mathcal{L}\{g(t)\}$$

__Ex:__ Find the Laplace transform of $\cosh(at)$ and $\sinh(at)$

Since $\cosh(at) = \dfrac{e^{at}+e^{-at}}{2}$ and $\sinh(at) = \dfrac{e^{at}-e^{-at}}{2}$

$$\mathcal{L}\{\cosh(at)\} = \mathcal{L}\left(\frac{e^{at}+e^{-at}}{2}\right) = \frac{1}{2}\mathcal{L}(e^{at}) + \frac{1}{2}\mathcal{L}(e^{-at})$$

$$= \frac{1}{2}\left(\frac{1}{s-a} + \frac{1}{s+a}\right) = \frac{s}{s^2-a^2} \quad \text{for } s>|a|$$

$$\mathcal{L}\{\sinh(at)\} = \mathcal{L}\left(\frac{e^{at}-e^{-at}}{2}\right) = \frac{1}{2}\mathcal{L}(e^{at}) - \frac{1}{2}\mathcal{L}(e^{-at})$$

$$= \frac{1}{2}\left(\frac{1}{s-a} - \frac{1}{s+a}\right) = \frac{a}{s^2-a^2}$$

for $s > |a|$

__Ex:__ Calculate the Laplace Transforms of $\cos(\omega t)$ and $\sin(\omega t)$

Let us denote the Laplace transform of $\cos(\omega t)$ by L_c and the Laplace transform of $\sin(\omega t)$ by L_s

Now $$L_c = \int_0^\infty e^{-st}\cos(\omega t)\,dt$$

$$= \frac{e^{-st}}{-s}\cos(\omega t)\Big|_0^\infty - \frac{\omega}{s}\int_0^\infty e^{-st}\sin(\omega t)\,dt$$

$$\Rightarrow L_c = \frac{1}{s} - \frac{\omega}{s}L_s \quad \ldots\ldots (1)$$

Now $$L_s = \int_0^\infty e^{-st}\sin(\omega t)\,dt$$

$$= -\frac{e^{-st}}{s}\sin(\omega t)\Big|_0^\infty + \frac{\omega}{s}\int_0^\infty e^{-st}\cos(\omega t)\,dt$$

$$= 0 + \frac{\omega}{s}L_c$$

$$\Rightarrow \quad L_s = \frac{\omega}{s} L_c \quad \ldots\ldots \text{②}$$

Solving ① and ②, $L_c = \frac{1}{s} - \frac{\omega^2}{s^2} L_c$

$$\Rightarrow \left(1 + \frac{\omega^2}{s^2}\right) L_c = \frac{1}{s}$$

$$\Rightarrow \frac{(s^2+\omega^2)}{s^2} L_c = \frac{1}{s} \Rightarrow \boxed{L_c = \frac{s}{s^2+\omega^2}}$$

Then ② $\Rightarrow$

$$L_s = \frac{\omega}{s} \frac{s}{s^2+\omega^2}$$

$$\Rightarrow \boxed{L_s = \frac{\omega}{s^2+\omega^2}}$$

Ex: Calculate the Laplace transform of t^{n+1}

$$\mathcal{L}(t^{n+1}) = \int_0^\infty e^{-st} t^{n+1} dt = -\frac{e^{-st}}{s} t^{n+1}\Big|_0^\infty + \frac{n+1}{s}\int_0^\infty e^{-st} t^n dt$$

$$= 0 + \frac{n+1}{s}\mathcal{L}(t^n) = \frac{n+1}{s}\mathcal{L}(t^n)$$

$$= \frac{(n+1)}{s}\frac{n}{s}\mathcal{L}(t^{n-1})$$

$$= \frac{(n+1)}{s}\frac{n}{s}\cdots\frac{2}{s}\frac{1}{s}\mathcal{L}(t^0)$$

$$= \frac{(n+1)}{s}\,\frac{n}{s}\cdots\frac{2}{s}\,\frac{1}{s}\cdot\frac{1}{s}$$

$$= \boxed{\frac{(n+1)!}{s^{n+2}}}$$

- **Some functions $f(t)$ and their Laplace Transform $\mathcal{L}(f)$**

(1) $f(t) = 1 \Rightarrow \mathcal{L}(f) = \frac{1}{s}$

(2) $f(t) = t \Rightarrow \mathcal{L}(f) = \frac{1}{s^2}$

(3) $f(t) = t^2 \Rightarrow \mathcal{L}(f) = \frac{2!}{s^3}$

(4) $f(t) = t^n \Rightarrow \mathcal{L}(f) = \frac{n!}{s^{n+1}}$
$(n = 0, 1, \ldots)$

(5) $f(t) = t^a \Rightarrow \mathcal{L}(f) = \frac{\Gamma(a+1)}{s^{a+1}}$
$(a > 0)$

(6) $f(t) = e^{at} \Rightarrow \mathcal{L}(f) = \frac{1}{s-a}$

(7) $f(t) = \cos \omega t \Rightarrow \mathcal{L}(f) = \frac{s}{s^2+\omega^2}$

(8) $f(t) = \sin \omega t \Rightarrow \mathcal{L}(f) = \frac{\omega}{s^2+\omega^2}$

(9) $f(t) = \cosh(at) \Rightarrow \mathcal{L}[f] = \dfrac{s}{s^2 - a^2}$

(10) $f(t) = \sinh(at) \Rightarrow \mathcal{L}(f) = \dfrac{a}{s^2 - a^2}$

(11) $f(t) = e^{at}\cos(\omega t) \Rightarrow \mathcal{L}(f) = \dfrac{(s-a)}{(s-a)^2 + \omega^2}$

(12) $f(t) = e^{at}\sin(\omega t) \Rightarrow \mathcal{L}(f) = \dfrac{\omega}{(s-a)^2 + \omega^2}$

Note:

(5) $\mathcal{L}(t^a) = \displaystyle\int_0^\infty e^{-st} t^a \, dt$

Let $st = x$
Then $t = \dfrac{x}{s}$
$dt = \dfrac{dx}{ds}$

$$= \int_0^\infty e^{-x} \left(\frac{x}{s}\right)^a \frac{dx}{s}$$

$$= \frac{1}{s^{a+1}} \int_0^\infty e^{-x} x^a \, dx \quad (\text{for } s > 0)$$

$$= \frac{\Gamma(a+1)}{s^{a+1}}$$

s-Shifting:

First Shifting Theorem:

If $f(t)$ has Laplace transform $F(s)$,
(where $s > k$ for some k)

then $e^{at} f(t)$ has the transform $F(s-a)$ (where $s-a > k$)

Thus $\mathcal{L}\{e^{at} f(t)\} = F(s-a)$

and $\mathcal{L}^{-1}\{F(s-a)\} = e^{at} f(t)$

<u>Proof:</u> $F(s-a) = \int_0^\infty e^{-(s-a)t} f(t)\,dt$

$$= \int_0^\infty e^{-st}\left(e^{at} f(t)\right)dt = \mathcal{L}\{e^{at} f(t)\}$$

(If $F(s)$ exists for some $s > k$, the first integral exists for $s-a > k$)

Now by taking the inverse Laplace transform we get $\mathcal{L}^{-1}\{F(s-a)\} = e^{at} f(t)$

<u>Ex:</u> Since $\mathcal{L}\{\cos(\omega t)\} = \dfrac{s}{s^2+\omega^2}$

and $\mathcal{L}\{\sin(\omega t)\} = \dfrac{\omega}{s^2+\omega^2}$

applying the shifting theorem,

$$\mathcal{L}\{e^{at}\cos(\omega t)\} = \boxed{\frac{(s-a)}{(s-a)^2+\omega^2}}$$

and $$\mathcal{L}\{e^{at}\sin(\omega t)\} = \frac{\omega}{(s-a)^2+\omega^2}$$

Ex: Find f if $\mathcal{L}(f) = \dfrac{2s-1}{s^2-6s+18}$

$$\mathcal{L}^{-1}\left\{\frac{2s-1}{s^2-6s+18}\right\} = \mathcal{L}^{-1}\left\{\frac{2(s-3)+5}{(s-3)^2+9}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{2(s-3)}{(s-3)^2+9}\right\} + \mathcal{L}^{-1}\left\{\frac{5}{(s-3)^2+9}\right\}$$

$$= 2\mathcal{L}^{-1}\left\{\frac{(s-3)}{(s-3)^2+3^2}\right\} + \frac{5}{3}\mathcal{L}^{-1}\left\{\frac{3}{(s-3)^2+3^2}\right\}$$

$$= 2e^{3t}\cos(3t) + \frac{5}{3}e^{3t}\sin(3t)$$

$$= \boxed{e^{3t}\left[2\cos(3t) + \frac{5}{3}\sin(3t)\right]}$$

Existence and Uniqueness of Laplace Transform:

If $f(t)$ is defined and is piecewise continuous on every finite interval on $t \geq 0$ and $|f(t)| \leq Me^{kt}$

for some constants M and K

(This is called the growth condition:
It means it doesnot grow too fast)

then the Laplace transform $\mathcal{L}(f)$ exists for all $s > k$ and if it exists, it is uniquely determined.

Conversely if two functions (both defined on the positive real axis) have the same transform, these functions cannot differ over an interval of positive length, although they may differ at isolated points. Thus inverse of a given transform is essentially unique.

In particular if two continuous functions have the same transform, they are completely identical.

<u>Proof for existence:</u>

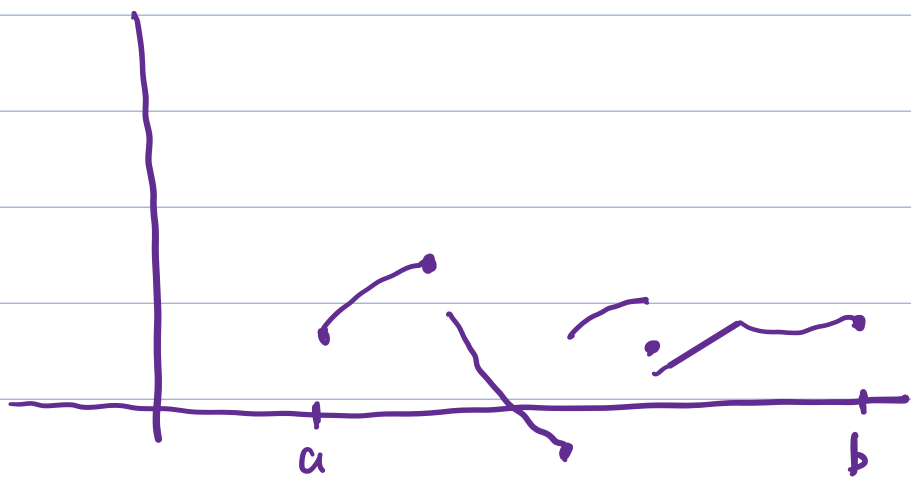
$$|\mathcal{L}(f)| = \left| \int_0^\infty e^{-st} f(t)\,dt \right| \leq \int_0^\infty |e^{-st}|\,|f(t)|\,dt$$

$$\leq \int_0^\infty M e^{kt} e^{-st}\,dt = M \int_0^\infty e^{-(s-k)t}\,dt$$

$$= M \left[-\frac{e^{-(s-k)}}{(s-k)} \right]_0^\infty = \frac{M}{s-k} < \infty$$

(for $s > k$).

Note: $f(t)$ is said to be piecewise continuous on $[a, b]$ if this interval can be divided into finitely many subintervals in each of which f is continuous and has finite limit as t approaches either endpoint of such subinterval from the interior. This then gives finite jumps as the only possible discontinuities.



Laplace Transforms of Derivatives and Integrals

Laplace Transform of first derivative:

If $f(t)$ is continuous for all $t \geq 0$ and satisfies the growth condition and $f'(t)$ is piecewise continuous. on every finite interval on the semiaxis $t \geq 0$, then

$$\boxed{\mathcal{L}(f') = s\mathcal{L}(f) - f(0)}$$

Proof: $\mathcal{L}(f')(s) = \int_0^\infty e^{-st} f'(t)\,dt$

$$= e^{-st} f(t)\Big|_0^\infty + s\int_0^\infty e^{-st} f(t)\,dt$$

$$= -f(0) + s\mathcal{L}(f)(s)$$

(The first expression is 0 at the upper limit when $s > k$)

Hence $\boxed{\mathcal{L}(f') = s\mathcal{L}(f) - f(0)}$

Laplace Transform of Second derivative:

If $f(t)$ and $f'(t)$ are continuous for all $t \geq 0$, satisfies the growth condition and f'' is piecewise continuous on every finite interval on the semiaxis $t \geq 0$, then

$$\boxed{\mathcal{L}\{f''\} = s^2\mathcal{L}\{f\} - sf(0) - f'(0)}$$

<u>Proof:</u> Applying the previous result

$$\mathcal{L}\{f''\} = s\mathcal{L}\{f'\} - f'(0)$$

$$= s(s\mathcal{L}\{f\} - f(0)) - f'(0)$$

$$\Rightarrow \boxed{\mathcal{L}\{f''\} = s^2\mathcal{L}\{f\} - sf(0) - f'(0)}$$

<u>Laplace Transform of the Derivative $f^{(n)}$ of any order:</u>

If $f, f', \ldots, f^{(n-1)}$ are continuous for all $t \geq 0$ and satisfy the growth condition and $f^{(n)}$ is piecewise continuous on every finite interval on the semiaxis $t \geq 0$, then

$$\boxed{\mathcal{L}\{f^{(n)}\} = s^n\mathcal{L}\{f\} - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - f^{(n-1)}(0)}$$

<u>Ex:</u> Let $f(t) = t\sin(\omega t)$ Find $\mathcal{L}\{f\}$

$$f(t) = t\sin(\omega t) \Rightarrow f(0) = 0$$

$$f'(t) = \sin(\omega t) + t\omega\cos(\omega t) \Rightarrow f'(0) = 0$$

$$f''(t) = \omega\cos(\omega t) + \omega\cos(\omega t) - t\omega^2\sin(\omega t)$$

$$\Rightarrow f''(t) = 2\omega\cos(\omega t) - \omega^2 t\sin(\omega t)$$

Now $\mathcal{L}\{f''\} = s^2\mathcal{L}\{f\} - s f(0) - f'(0) = s^2\mathcal{L}\{f\}$

$$\mathcal{L}\{f''\} = \mathcal{L}\{2\omega\cos(\omega t) - \omega^2 t\sin(\omega t)\}$$

$$= 2\omega\frac{s}{s^2+\omega^2} - \omega^2\mathcal{L}\{f\}$$

$$\Rightarrow \quad s^2\mathcal{L}\{f\} = \frac{2\omega s}{s^2+\omega^2} - \omega^2\mathcal{L}\{f\}$$

$$\Rightarrow \quad (s^2+\omega^2)\,\mathcal{L}\{f\} = \frac{2\omega s}{s^2+\omega^2}$$

$$\Rightarrow \quad \boxed{\mathcal{L}\{f\} = \frac{2\omega s}{(s^2+\omega^2)^2}}$$

<u>Ex:</u> <u>Another way of calculating $\mathcal{L}\{\cos(\omega t)\}$ and $\mathcal{L}\{\sin(\omega t)\}$:</u>

Let $\quad f(t) = \cos(\omega t)$

$f(0) = 1$, $f'(t) = -\omega\sin(\omega t) \Rightarrow f'(0) = 0$

$f''(t) = -\omega^2\cos(\omega t)$

$$\mathcal{L}(f'') = s^2\mathcal{L}(f) - s f(0) - f'(0)$$

$$\mathcal{L}\{-\omega^2\cos(\omega t)\} = s^2\mathcal{L}\{\cos(\omega t)\} - s\times 1 - 0$$

$$\Rightarrow -\omega^2\mathcal{L}\{\cos(\omega t)\} = s^2\mathcal{L}\{\cos(\omega t)\} - s$$

$$\Rightarrow \quad (s^2+\omega^2)\mathcal{L}\{\cos(\omega t)\} = s$$

$$\Rightarrow \quad \boxed{\mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2+\omega^2}}$$

Similarly let $g(t) = \sin(\omega t)$

$g(0)=0$, $g'(t) = \omega\cos(\omega t) \Rightarrow g'(0) = \omega$

$g''(t) = -\omega^2\sin(\omega t)$

Now $\mathcal{L}\{g''\} = s^2\mathcal{L}\{g\} - sg(0) - g'(0)$

$$\Rightarrow \mathcal{L}\{-\omega^2\sin(\omega t)\} = s^2\mathcal{L}\{\sin(\omega t)\} - s\times 0 - \omega$$

$$\Rightarrow -\omega^2\mathcal{L}\{\sin(\omega t)\} = s^2\mathcal{L}\{\sin(\omega t)\} - \omega$$

$$\Rightarrow \quad (s^2+\omega^2)\mathcal{L}\{\sin(\omega t)\} = \omega$$

$$\Rightarrow \boxed{\mathcal{L}\{\sin(\omega t)\} = \frac{\omega}{(s^2+\omega^2)}}$$

Laplace Transform of Integral:

If $f(t)$ is a piecewise continuous function for all $t \geq 0$ and satisfies the growth condition then for $s > k > 0$ and $t > 0$ we have

$$\mathcal{L}\left\{\int_0^t f(\tau)\,d\tau\right\} = \frac{1}{s}F(s) \quad \left(= \frac{1}{s}\mathcal{L}\{f\}(s)\right)$$

$$\left(\text{Therefore} \quad \int_0^t f(\tau)\,d\tau = \mathcal{L}^{-1}\left\{\frac{1}{s}F(s)\right\}\right)$$

Proof: Since $f(t)$ is piecewise continuous, it integral is continuous and it satisfies the growth condition because

$$\left|\int_0^t f(\tau)\,d\tau\right| \leq \int_0^t |f(\tau)|\,d\tau \leq \int_0^t Me^{k\tau}\,d\tau$$

$$= \frac{M\left(e^{kt}-1\right)}{k} \leq \frac{Me^{kt}}{k}$$

Also the derivative of the integral is $f(t)$ except at points of discontinuity of f.

Applying the first derivative theorem to the integral of f we get

$$\mathcal{L}\left\{\left(\int_0^t f(\tau)\,d\tau\right)'\right\}(s) = s\,\mathcal{L}\left\{\int_0^t f(\tau)\,d\tau\right\} - \int_0^0 f(\tau)\,d\tau$$

$$\Rightarrow \mathcal{L}\{f(t)\}(s) = s\,\mathcal{L}\left\{\int_0^t f(\tau)\,d\tau\right\}$$

$$\Rightarrow \mathcal{L}\left\{\int_0^t f(\tau)\,d\tau\right\} = \frac{1}{s}\mathcal{L}\{f\}(s)$$

Ex: Find the inverse Laplace Transform of $\dfrac{1}{s(s^2+\omega^2)}$

and $\dfrac{1}{s^2(s^2+\omega^2)}$

Since $\mathcal{L}\{\sin(\omega t)\} = \dfrac{\omega}{s^2+\omega^2}$

we have $\mathcal{L}\left\{\dfrac{\sin(\omega t)}{\omega}\right\} = \dfrac{1}{s^2+\omega^2}$

$$\Rightarrow \mathcal{L}^{-1}\left\{\frac{1}{s(s^2+\omega^2)}\right\} = \int_0^t \frac{\sin \omega\tau}{\omega}\,d\tau = \left[-\frac{\cos(\omega\tau)}{\omega^2}\right]_0^t$$

$$= \boxed{\frac{[1-\cos(\omega t)]}{\omega^2}}$$

Now since $\mathcal{L}\left\{\dfrac{[1-\cos(\omega t)]}{\omega^2}\right\} = \dfrac{1}{s(s^2+\omega^2)}$, $\mathcal{L}^{-1}\left\{\dfrac{1}{s^2(s^2+\omega^2)}\right\} = \displaystyle\int_0^t \frac{[1-\cos \omega\tau]}{\omega^2}\,d\tau$

$$= \left[\frac{\tau}{\omega^2} - \frac{\sin(\omega\tau)}{\omega^3}\right]_0^t = \boxed{\left[\frac{t}{\omega^2} - \frac{\sin(\omega t)}{\omega^3}\right]}$$